Assignment 1 - Cat vs Dog Image Classification

1. Abstract

This report presents a binary image classification system for distinguishing cats and dogs using a pretrained ResNet-18 model. Two training strategies were evaluated: transfer learning and full fine-tuning. Under identical training settings, transfer learning achieved a validation accuracy of 97.93%, outperforming full fine-tuning (91.00%) while converging substantially faster. A FastAPI-based web application was also developed to support real-time inference. This report summarizes the dataset preparation process, methodology, experimental results, limitations, and potential future improvements.

2. Introduction

Cat and dog classification is a widely used benchmark problem in computer vision and image classification. With the availability of pretrained deep learning models, transfer learning has become an effective approach for achieving high performance while reducing training time and computational cost.

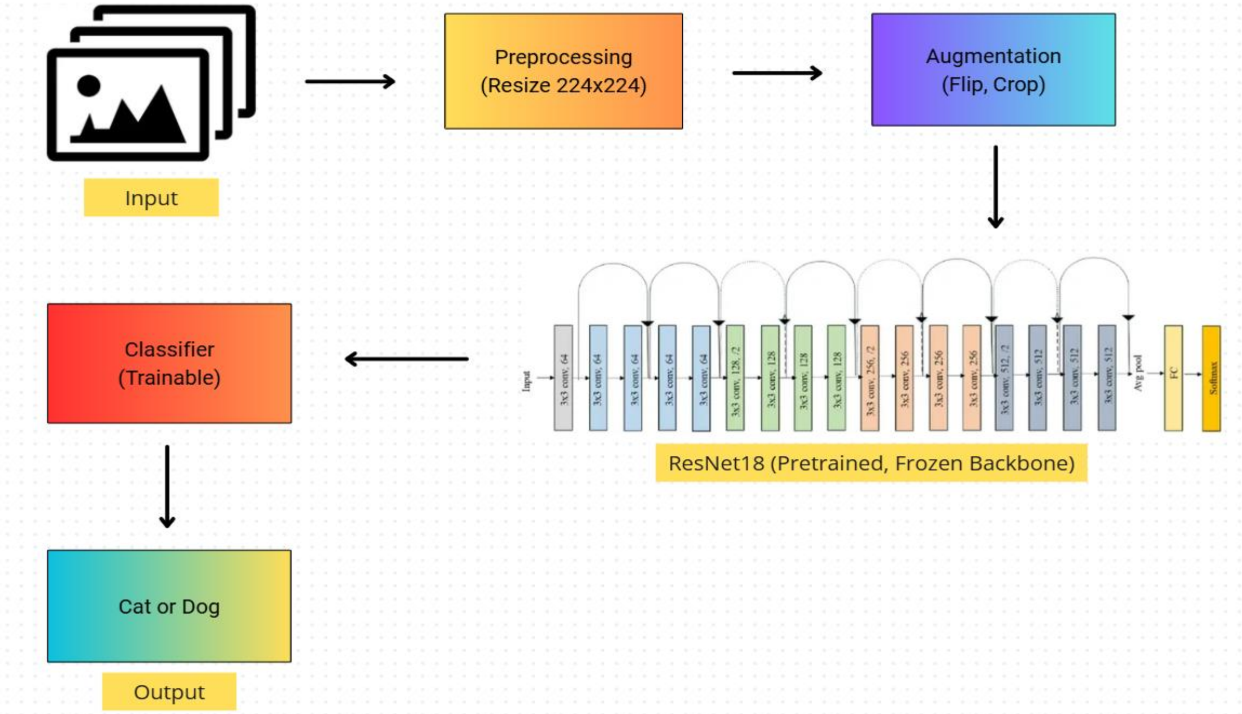
The objective of this project is to develop a binary image classification system capable of distinguishing between cats and dogs. To investigate different training strategies, both transfer learning and full fine-tuning were evaluated using a pretrained ResNet-18 model. In addition, a FastAPI-based web application was implemented to provide an interface for model inference

3. Methodology

3.1 Pipeline Overview

The overall classification pipeline is illustrated in the pipeline diagram. The workflow consists of the following stages:

1. Images are collected and split into training, validation, and test sets.
2. Preprocessing and normalization are applied to ensure consistent input dimensions and pixel distributions.
3. Data augmentation techniques, including random resized cropping and horizontal flipping, are applied during training to improve model generalization.
4. The transformed images are forwarded through a pretrained ResNet-18 backbone for feature extraction.
5. During transfer learning, the backbone parameters remain frozen while only the classification head is optimized on the target dataset.
6. The classification head produces the final prediction, classifying the input image as either Cat or Dog



3.2 Data Augmentation

To improve model generalization and reduce overfitting, data augmentation techniques, including Random Resized Crop and Horizontal Flip, were applied during training. In addition, all images were normalized using ImageNet statistics before being fed into the model.

3.3 Model Architectures

ResNet-18 pretrained on ImageNet was selected as the backbone network due to its strong performance and computational efficiency. To investigate the impact of different training strategies, two approaches were evaluated. In the transfer learning setting, the pretrained backbone was frozen and only the final classification head was optimized on the target dataset. In contrast, full fine-tuning updated all model parameters, including both the pretrained backbone and the classification head.

4. Experimental Results

To ensure a fair comparison between transfer learning and full fine-tuning, the same training configuration was applied to both approaches.

Paramters	**Value**
Backbone	ResNet-18
Optimizer	Adam
Learning Rate	1e-4
Maximum Epochs	30
Early Stopping	Patience = 5
Data Augmentation	Random Resized Crop, Horizontal Flip

Model selection was based on validation performance, and early stopping was employed to prevent overfitting and reduce unnecessary training epochs.

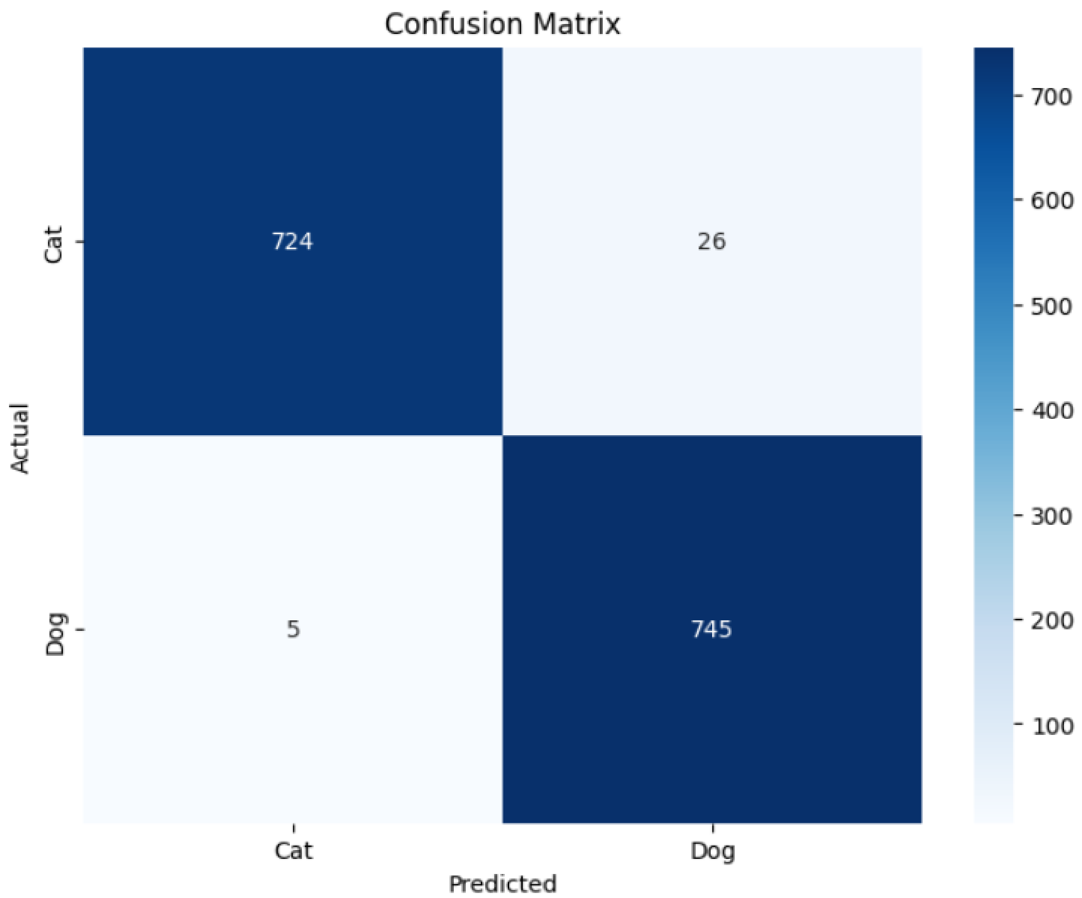
5. Results

The experimental results are summarized below.

Methods	**Accuracy(%)**	**F1-Score(%)**	**Stopped Epochs**
Transfer Learning	97.93	97.93	7
Full Fine-tuning	91.00	90.89	29

From the results, transfer learning consistently outperforms full fine-tuning in terms of both validation accuracy and F1-score. In addition, the transfer learning model converges significantly faster, stopping at epoch 7 compared to 29 epochs for full fine-tuning under the same training configuration. This indicates that leveraging pretrained ImageNet features provides a more efficient initialization for optimizing the Cat vs Dog classification task.

To further analyze the performance of the best-performing model, the confusion matrix of the transfer learning approach is shown below. The results indicate that the model achieves strong classification performance on both classes, with only a small number of misclassifications observed.



Overall, the experimental results demonstrate that transfer learning is more effective than full fine-tuning for this task in terms of both predictive performance and training efficiency

6. Remaining Problems

Although the proposed model achieved strong performance on the evaluation dataset, several limitations remain. First, the model was trained and evaluated exclusively on natural photographic images; therefore, its performance on other visual domains, such as cartoons, sketches, paintings, or AI-generated images, remains uncertain. Second, the dataset primarily contains images in which the target animal is the dominant object, and the model may experience performance degradation in more complex scenes involving cluttered backgrounds, multiple objects, or partial occlusions. In addition, challenging visual conditions, including low-light environments, extreme viewpoints, and motion blur, may negatively affect classification accuracy.

7. Future Improvements

Future improvements include enhancing data diversity and applying more advanced augmentation techniques to improve robustness under different visual conditions. Additionally, experimenting with stronger optimization strategies such as SGD with momentum or AdamW with weight decay may further improve generalization. Expanding the dataset with more diverse real-world images and introducing hard examples (e.g., occlusions, low-light, and multi-object scenes) could also help the model become more robust. Finally, extending the model to support open-set recognition would allow it to reject irrelevant inputs outside the cat–dog categories.

8. Conclusion

This report presented a binary image classification system for distinguishing cats and dogs using a pretrained ResNet-18 model. Two training strategies, transfer learning and full fine-tuning, were evaluated under the same experimental settings. Results showed that transfer learning significantly outperformed full fine-tuning in both accuracy and convergence speed.

In addition, a FastAPI-based web application was developed to enable real-time inference, demonstrating the practical applicability of the proposed system.

Assignment 2 - Text To Speech

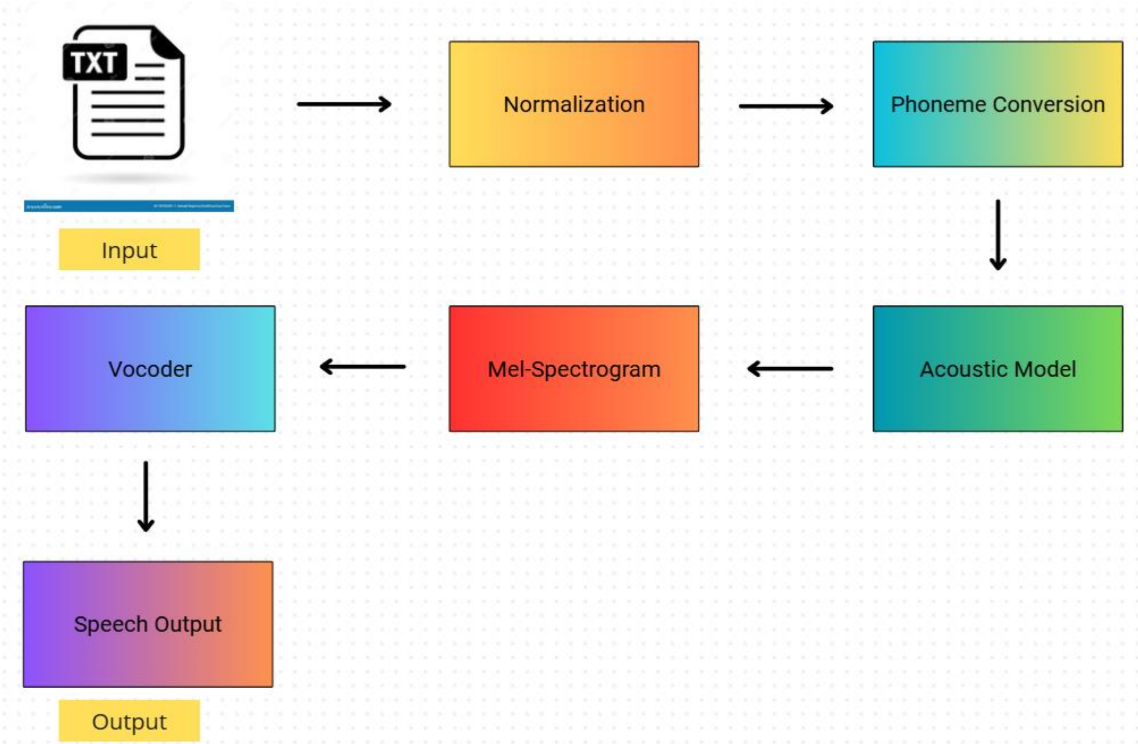
1. Introduction

Text-to-Speech (TTS) is a technology that converts written text into natural-sounding speech. It plays an important role in accessibility systems, virtual assistants, and human-computer interaction, especially for users with visual impairments or reading difficulties.

This report presents a proposed design for a Vietnamese Text-to-Speech system, describing the overall pipeline, key components, and potential challenges in building a high-quality speech synthesis model.

2. Proposed Pipeline

The proposed TTS system consists of three main stages: text processing, acoustic feature generation, and waveform synthesis. The input text is first normalized and converted into a phoneme or linguistic representation to better capture pronunciation information. This representation is then passed to an acoustic model that generates intermediate speech features such as Mel-spectrograms, which encode the time-frequency characteristics of speech. Finally, a neural vocoder converts these features into a natural-sounding audio waveform. The overall pipeline is illustrated in the diagram below.



3. System Components

For the acoustic model, modern neural TTS architectures such as Tacotron 2 or FastSpeech can be used to map text or phoneme sequences to Mel-spectrogram representations. For waveform synthesis, neural vocoders such as WaveGlow or HiFi-GAN can generate high-quality speech signals from spectrograms.

Vietnamese language processing requires a proper phonemization step due to the tonal nature of the language. Therefore, a grapheme-to-phoneme (G2P) module or rule-based phoneme conversion system is essential to improve pronunciation accuracy.

4. Challenges and Solutions

Building a Vietnamese TTS system introduces several challenges. First, Vietnamese is a tonal language, and incorrect tone modeling can significantly affect meaning. This can be addressed by incorporating tone-aware phoneme representations.

Second, high-quality paired text-audio datasets are limited, which may affect model generalization. This issue can be mitigated by data augmentation techniques or transfer learning from pre-trained multilingual TTS models.

Third, naturalness and prosody modeling remain difficult, especially for long sentences. Attention-based or duration-based models such as FastSpeech can help improve stability and reduce pronunciation errors.

5. Conclusion

This report proposed a neural Text-to-Speech system for Vietnamese language synthesis. The system is based on a modular pipeline consisting of text processing, acoustic modeling, and waveform generation. Key challenges such as tonal accuracy, data scarcity, and speech naturalness were discussed along with potential solutions. Future improvements may focus on leveraging larger datasets and more advanced speech synthesis architectures to improve audio quality and robustness.